

- 3 (a) (i) power = work done per unit time / energy transferred per unit time / rate of work done B1 [1]
- (ii) Young modulus = stress / strain B1 [1]
- (b) (i) 1. $E = T / (A \times \text{strain})$ (allow strain = ϵ) C1
 $T = E \times A \times \text{strain} = 2.4 \times 10^{11} \times 1.3 \times 10^{-4} \times 0.001$ M1
 $= 3.12 \times 10^4 \text{ N}$ A0 [2]
2. $T - W = ma$ C1
 $[3.12 \times 10^4 - 1800 \times 9.81] = 1800a$ C1
 $a = 7.52 \text{ ms}^{-2}$ A1 [3]
- (ii) 1. $T = 1800 \times 9.81 = 1.8 \times 10^4 \text{ N}$ A1 [1]
2. potential energy gain = mgh C1
 $= 1800 \times 9.81 \times 15$
 $= 2.7 \times 10^5 \text{ J}$ A1 [2]
- (iii) $P = Fv$ C1
 $= 1800 \times 9.81 \times 0.55$ C1
input power = $9712 \times (100/30) = 32.4 \times 10^3 \text{ W}$ A1 [3]

- 4 (a) p.d. = energy transformed from electrical to other forms